Designing AI-Assisted Co-Writing for Tribal Government Emergency Operations Planning.

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Abstract. Emergency Operations Plans (EOPs) are critical for disaster preparedness and response, yet many tribal emergency managers face significant challenges in translating operational knowledge into comprehensive written plans. This study presents the design and evaluation of AI-Inscriptior, an AI-assisted co-writing tool developed to support (U.S Tribal Nation governments and their) Tribal Emergency Operations Plan. The main purpose of the system is to guide the users through and generate the draft policy text using the standardized EOP structures, thereby reducing the documentation burden and supporting iterative refinements through targeted prompts. The application helped to develop core EOP chapters and hazard-specific annexes such as wildfire, flood, and extreme weather scenarios commonly relevant to Tribal communities, illustrating the capability of AI to organize and articulate emergency management knowledge considering local context and decision-making authority. Our study generated qualitative insights by conducting interviews with emergency management professionals and facilitating a panel discussion with national tribal stakeholders to capture diverse perspectives on usability, trust, and real-world applicability of the system. The findings indicate that AI assisted co-writing can substantially accelerate and improve the quality and agility of EOP development by reducing the blank-page burden and improving consistency across plan sections, provided it is used as a complementary tool to expert judgment while preserving tribal sovereignty, cultural sensitivity, and environmental considerations. Drawing on these findings, we articulate design implications that surface current limitations of the system and outline future directions for strengthening AI support for Tribal emergency management practices in real-world disaster response scenarios.

Keywords. Data Sovereignty, AI-assisted Co-writing, emergency operations plans, tribal governance, disaster management

1. Introduction

In the past decade alone, over 70 natural disasters have struck Tribal communities, with some nations experiencing multiple nature diaster (Herrera, 2019; Manson and Buchwald, 2021). Frequent and severe disasters on Tribal lands have strengthen the need for robust emergency planning, yet Tribal governments face unique challenges in meeting federal documentation requirements. To access federal disaster assistance, Tribes must produce detailed Emergency Operations Plans (EOPs) and damage reports aligned with Federal Emergency Management Agency (FEMA)

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guidelines. This process is often burdensome and complex: developing and maintaining these plans can be costly and time-consuming, and conducting damage assessments taxes limited local resources (Tsai et al., 2024). As a result, many Tribal governments have historically relied on state-issued disaster declarations instead of declaring emergencies themselves, a compromise that can diminish Tribal sovereignty and limit direct funding. These challenges point to an urgent need for tools that can streamline documentation without sacrificing local control (Tsai et al., 2024).

At the same time, advances in generative artificial intelligence (AI) offer new opportunities to assist with governmental writing tasks and improve public-sector efficiency. Large Language Models (LLMs) such as GPT-3.5 and GPT-4 have demonstrated the ability to generate well-structured text, prompting exploration of their use in domain-specific documentation. For example, FEMA recently piloted an LLM-based assistant to help local governments draft Hazard Mitigation Plans, aiming to make communities more resilient and speed up grant eligibility (Security, 2025). Early evidence in broader public administration suggests that generative AI can accelerate bureaucratic writing; one quasi-experimental study found that a tailored AI system reduced drafting time for government officials and particularly helped new employees by guiding them through formal document structures (Kim, 2026). Likewise, sectors such as law and healthcare are experimenting with AI-driven drafting tools to increase access and consistency in documentation (Satterfield, 2025). However, deploying AI in high-stakes government contexts also raises concerns about accuracy, trust, and ethics. Users worry about AI "hallucinations" (plausible-sounding but incorrect content) and the need to verify AI outputs, which can offset efficiency gains (Tsai et al., 2024; Gupta et al., 2024). Moreover, in Tribal contexts, any technological intervention must respect Indigenous data sovereignty and cultural knowledge. Tribal leaders emphasize that AI adoption should not repeat colonial patterns of extraction; data used to train or run models must be governed under Tribal authority to avoid misrepresentation or misuse of Indigenous knowledge (Rainie et al., 2019). These considerations mean that an AI tool for Tribal emergency planning must be carefully designed to function as a supportive co-writer—accelerating tedious documentation tasks—while leaving all judgment and decision-making in human hands.

Prior research on generative AI has largely focused on general-purpose tools and their use in broad contexts such as academic writing and information generation (Cheng et al., 2025). While these studies highlight important concerns, including hallucination, accuracy, and ethical use, they provide limited insight into how AI systems operate in domain-specific, high-stakes settings. Emerging work in public administration has begun to examine the use of large language models, such as ChatGPT, to improve bureaucratic productivity (Kim, 2026). However, this line of research primarily emphasizes efficiency and task-level performance, with less attention to how these systems are experienced, interpreted, and integrated within governance-sensitive, domain-specific contexts. In particular, there is a lack of research on AI systems designed for localized use in settings such as tribal emergency management, where considerations of data sovereignty, trust, and contextual alignment are critical.

This study investigates an interdisciplinary solution at the intersection of AI, emergency management, and Tribal governance. We present AI Inscriptor, an AI-assisted co-writing system tailored for developing Tribal EOPs in alignment with FEMA guidelines. The system is designed as a collaborative tool that augments emergency managers' capabilities rather than replacing their judgment. Built on an open-sourced LLM, AI Inscriptor provides structured prompts and draft text suggestions for each section of an EOP (e.g. Purpose, Concept of Operations, hazard-specific annexes), guiding users through the complex documentation requirements. A key feature of our approach is an emphasis on Tribal sovereignty and trust: all AI processing runs on secure, locally hosted models, so that no data ever leaves the Tribal or institutional environment. This local deployment directly addresses the privacy and control issues identified with cloud-based AI services, ensuring that data stays local and under Tribal ownership – an important baseline for trust as echoed in community feedback (Tsai et al., 2024). Furthermore, the system's prompts are carefully crafted to elicit community-specific details (e.g. Tribal roles, cultural priorities, mutual aid relationships) within the FEMA-recommended template. By doing so, the AI-generated content serves as a starting point that reflects federal format expectations yet remains flexible for local knowledge and cultural correctness. Our goal is to reduce the documentation workload – potentially compressing months of writing into weeks – while keeping Tribal experts in the loop of plan content and decision-making.

Specifically, we aim to answer the research question: *How do tribal practitioners perceive and respond to an AI assisted co writing tool for tribal emergency operations planning?* To answer this question, we conducted a qualitative, design oriented study with Tribal emergency management practitioners. Using a mixed-method approach (document analysis, system prototyping, and user evaluations), we examined how participants interacted with AI Inscriptor and how it affected their planning workflow. Through in-depth expert interviews and a panel discussion with broader Tribal stakeholders, we gathered insights on usability, perceived trustworthiness, and the sys-

tem's alignment with both FEMA requirements and Tribal values. The findings reveal that AI assistance can indeed accelerate EOP development and improve clarity of documentation. However, our results also indicate that speed alone is not sufficient for real-world adoption. Issues of trust, legitimacy, and control emerged as deciding factors: users valued the AI's help in structuring and drafting content but only if the content remained accurate, locally relevant, and subject to human review. Participants stressed that an AI planning tool must function as a supportive role rather than an authoritative planner, meaning that final decisions and policy judgments stay firmly in human hands. Moreover, the panel discussion highlighted broader considerations of sustainability and self-determination – for example, concerns that heavy reliance on AI could introduce new labor burdens, environmental costs, and questions about how such innovations align with long-term Tribal capacity building. These nuanced perspectives inform a set of design and governance considerations for responsible AI integration in the public sector.

This study addresses this gap by examining a case study of a generative AI tool applied to a domain-specific, high-stakes task: emergency operations plan writing in tribal governance. The contributions are two-fold: (1) a novel co-writing system that demonstrates how collaborative intelligence between human experts and AI can streamline the creation of complex emergency plans; and (2) an in-depth evaluation of this system in a Tribal context, yielding insights on user experience, trust dynamics, and the sociotechnical factors that shape AI adoption in public institutions. By situating our research in the context of Tribal emergency management, we also extend the discourse on digital transformation for public value – illustrating how technologies like AI can create public value (e.g. more timely, compliant plans) only if designed with inclusivity, cultural resonance, and accountability in mind. The next sections detail the related work that informed our approach, followed by the system design and methodology, and then the findings and implications for digital government research and practice.

2. Literature Review

2.1. AI in Digital Government and Document Automation

Governments worldwide are exploring generative AI to improve writing-intensive processes, from drafting correspondence to formulating policy documents. The appeal is clear: AI can help public agencies produce clearer, more accessible text and reduce manual workload (GovTech, 2023; Blomkamp, 2018). For instance, recent initiatives have used LLMs to translate bureaucratic jargon into plain language in citizen communications. Early evidence suggests significant efficiency gains – in one case, public servants using ChatGPT to draft routine emails and reports found it sped up their work, albeit with careful oversight to avoid errors. However, the public sector's adoption of AI writing assistants comes with strict requirements around security, accuracy, and accountability. Agencies are urged to keep sensitive data in-house and not rely on consumer AI platforms that send data to the cloud. The importance of maintaining control over data and models is often highlighted in official guidance and executive directives. Additionally, because government documents carry legal and public trust implications, generated text must be verifiable and fit-for-purpose. Research in human–AI collaboration underscores that AI is best used to augment human writers rather than fully automating content creation (Luger and Sellen, 2016; Jonas and Hanrahan, 2022). Our work aligns with these principles by using a locally hosted LLM and positioning the AI as a co-writer whose outputs are always subject to human review and editing.

2.2. AI-Assisted Writing in Domain-Specific Contexts

Beyond general communications chatbots (Adamopoulou and Moussiades, 2020; Almansor and Hussain, 2020), domain-specific applications of AI writing support are emerging in fields like healthcare (Song et al., 2025), law (Han et al., 2024), and education Kim (2026) . These studies provide analogies for our approach in emergency management. Notably, in clinical settings AI assistants have been developed to help write medical documentation such as patient discharge summaries. Song et al. (2025) introduced an on-site LLM-based assistant for emergency department doctors; it generated draft discharge notes that physicians could then edit (Song et al., 2025). The results were striking – the AI-assisted notes were judged more complete and concise than the purely manual notes, and the median writing time was cut by over 50%. Crucially, this system was deployed within the hospital's own secure environment, ensuring patient data never left local servers, which increased clinicians' trust in using the tool. Similar trust-driven design choices are seen in legal and business domains, where proprietary LLM solutions (e.g. for drafting contracts or summarizing reports) emphasize data confidentiality and the ability for users to post-edit the output (Regalia, 2024). These domain-specific successes suggest that AI co-writing tools excel when they are tailored to a specific structure and jargon, allowing them to integrate smoothly into professional workflows. In our case, the FEMA-aligned EOP template and emergency management terminology serve as this structural backbone and vocabulary. By training prompts and system behavior around this domain knowledge, we aim to achieve a comparable boost in efficiency and quality for emergency planning documents.

2.3. AI in Emergency Management and Disaster Planning

Within the disaster research community, artificial intelligence has predominantly been applied to quantitative and decision-support challenges – for example, using machine learning for hazard prediction, geospatial and remote sensing approaches for damage assessment, and AI-based systems for crisis information management (Wibowo et al., 2025). A recent bibliometric review by Wibowo et al. (2025), based on 848 publications, identified major clusters of AI research in disaster management focused on areas such as disaster monitoring and prediction, geospatial risk analysis, decision support systems, and social media-based response analysis, reflecting a strong emphasis on analytics and automation in response efforts(Wibowo et al., 2025). Far less explored is the use of AI to support the planning and administrative side of emergency management, such as developing plans, checklists, and after-action reports (Ishiwatari et al., 2023). Emergency planning remains a domain where human expertise, institutional knowledge, and compliance requirements intersect in complex ways that have not yet been fully addressed by technology (Barnosky et al., 2022). Traditional planning guidance (e.g. FEMA's CPG 101) provides frameworks but still relies on substantial human labor to document and integrate information from diverse stakeholders (Federal Emergency Management Agency, 2023; Cordova, 2022). Some initial work is now bridging this gap: for example, Lee et al. (2024) introduced PlanAID (Lee et al., 2024), a prototype that leverages large language models to support emergency planning and decision-making tasks. The system assists users in framing planning problems and exploring potential response strategies in an interactive manner. This indicates a growing interest in applying AI not only for disaster response analytics but also for preparedness and planning activities(Tsai et al., 2023, 2019). Our research builds on this momentum by focusing on EOP documentation – a cornerstone of preparedness – and doing so through a co-creative paradigm (AI helping to write the plan text, guided by human inputs). In contrast to PlanAID's goal of optimizing plan logic, our solution's goal is to streamline plan writing while preserving the human-centered, narrative aspects of planning that ensure the resulting document is usable and tribal locally legitimate. This complements other tech-enabled approaches in emergency management (e.g. collaborative online planning platforms) and pushes the boundary on how intelligent tools can reduce the known gap between "plans on paper" and operational reality.

2.4. Indigenous Perspectives and Data Sovereignty

A critical dimension of our work is the Tribal context – namely, how AI tools can be introduced in Indigenous governance settings responsibly. Prior research and advocacy highlight that technology in Indigenous communities must respect sovereignty, self-determination, and cultural values at every step (Policy Options, 2025; Abdelnour-Nocera et al., 2013; Burnette et al., 2014). This includes adhering to Indigenous data sovereignty principles, which assert that communities should control their data and how it's used (Carroll et al., 2019; Kukutai and Taylor, 2016; Walter et al., 2021). AI initiatives have sometimes clashed with these principles when external systems harvest Indigenous knowledge or language data without consent. The OCAP principles (Ownership, Control, Access, Possession) were formulated to guard against such digital colonialism, insisting that Indigenous peoples retain ownership and governance over their information (Lewis et al., 2020; National Congress of American Indians, 2018). In the public sector context, this requires that AI systems be evaluated not only for technical performance but also for governance alignment, including questions of ownership of AI-generated outputs (e.g., drafts of Tribal EOPs), the storage and processing of sensitive emergency planning data, and the authority to audit, regulate, or discontinue system use. Our approach heeds calls from Indigenous technology scholars who argue that communities should move from being mere subjects of tech deployment to co-creators and decision-makers in how AI is applied (Parsons et al., 2016; Peters et al., 2018; Simonsen and Robertson, 2013).. Concretely, by involving Tribal partners in the design of AI Inscriptor and ensuring all data stays under Tribal jurisdiction, we aim to model a form of "Indigenous-led AI" development. This approach also aligns with broader movements in digital government toward accountable, inclusive digital ecosystems and collaborative digital transformation that centers public values. Our related work review thus underlines that any application of generative AI in government – especially in Indigenous and high-stakes contexts – must integrate technical innovation with ethical, cultural, and governance considerations. These insights set the stage for our methodology, where we describe how we operationalized these principles in the system design and research process.

3. Methodology

This study adopts a qualitative, design-oriented methodology to develop and evaluate an AI-assisted co-writing system for Tribal Emergency Operations Planning (EOP). The methodology integrates document analysis, system design, and applied testing to examine how generative AI can support the production of FEMA-aligned emergency documentation while preserving Tribal sovereignty and local control.

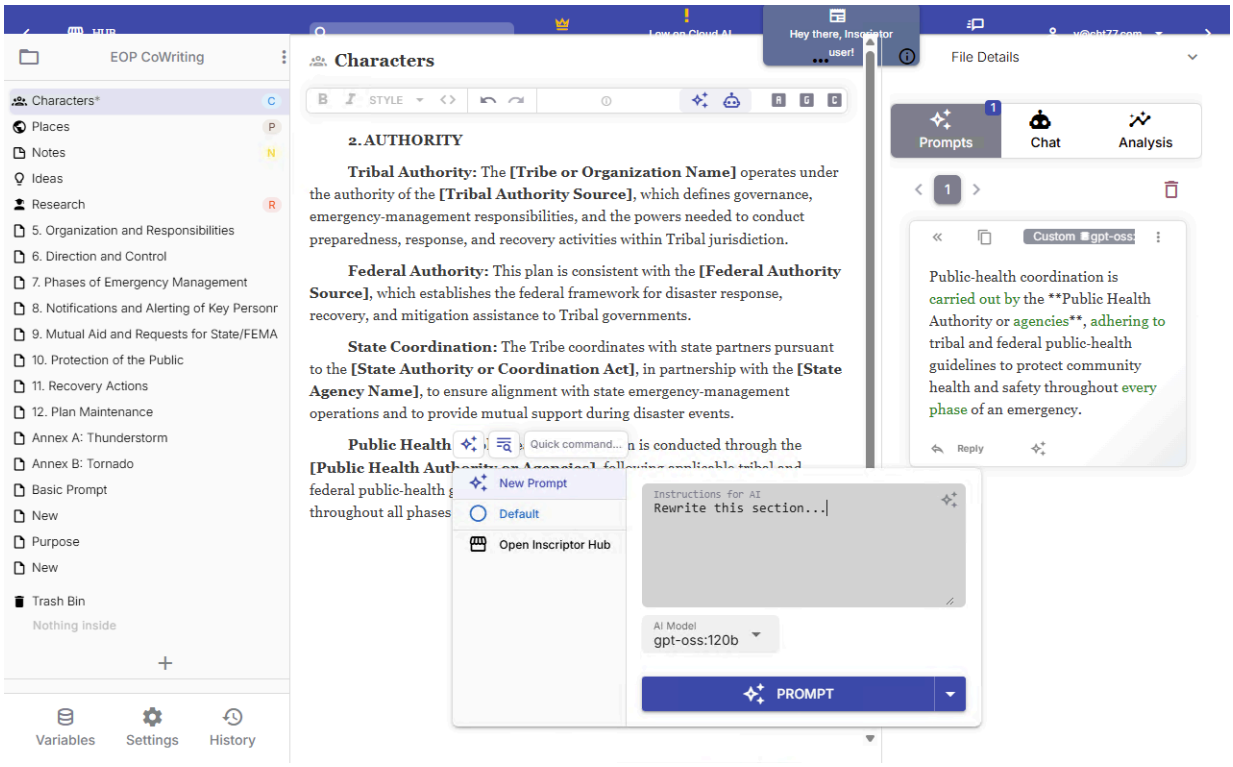


Fig. 1 – Inscriptor interface showing the structured Emergency Operations Plan (EOP) outline, chapter-based editing workspace, and AI-assisted text generation panel with predefined prompts for rapid draft creation and refinement.

3.1. Document Analysis and Requirements Identification

Document analysis was conducted to ground system design in the regulatory, operational, and governance contexts of Tribal emergency management. FEMA's Tribal Declarations Pilot Guidance and related emergency documentation requirements were reviewed to identify mandatory elements for federal assistance eligibility, including EOP activation criteria, documentation of exhausted Tribal resources, damage assessment narratives, and adherence to standardized reporting structures (Federal Emergency Management Agency, 2023). This analysis highlighted points in the planning and documentation process where administrative burden and technical complexity are most pronounced. In parallel, an existing Emergency Operations Plan from a Midwestern Tribal Nation—the Omaha Tribe of Nebraska—was analyzed to understand how federal requirements are implemented within a Tribal governance framework. The document was drafted and approved in 2011 by the Tribal Council. Our analysis focused on EOP structure (e.g., planning process, risk assessment, mitigating strategy, and plan maintenance), functional annexes, role definitions, hazard-specific planning, and the incorporation of Tribal values and decision-making practices. This document analysis identifies FEMA's required elements and the real world template that yields a council approved document. The analysis's conclusions will be employed to steer the preparation of AI assisted co-writing tools.

3.2. System Design: AI Inscriptor

Based on the identified requirements, the study employed Inscriptor, an AI-enabled writing platform [1], as the foundation for system development. Inscriptor integrates customizable AI tools designed to support brainstorming, drafting, refinement, and analysis of written content. For this study, Inscriptor was configured as a co-writing environment, explicitly positioned as a decision-support tool rather than an automated planning system. A FEMA-aligned Emergency Operations Plan template, derived from the analyzed Tribal EOP, was created and embedded within the Inscriptor platform. This template defines the overall structure of the document, including core chapters and annexes, and serves as a scaffold for user interaction. Emergency managers interact with the system by selecting specific EOP sections—such as Purpose and Scope, Situation and Assumptions, Organization and Responsibilities, or Recovery—and generating content within those predefined boundaries.

[1] https://inscriptor.io/

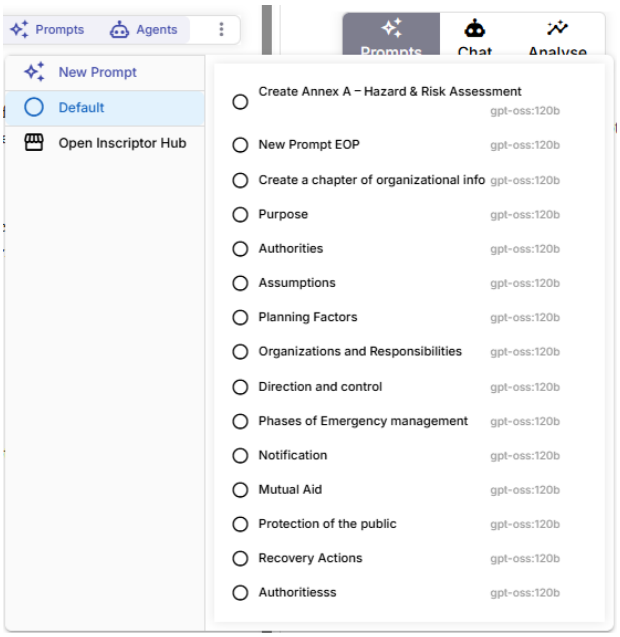


Fig. 2 – Structured prompt selection interface in AI Inscriptor. Users select predefined FEMA-aligned prompt templates corresponding to specific EOP chapters and hazard-specific annexes, enabling guided and consistent content generation.

Figure 1 illustrates the Inscriptor user interface. The left-hand sidebar displays the hierarchical structure of the Emergency Operations Plan (EOP), including core chapters—such as Organization and Responsibilities—as well as hazard-specific annexes (e.g., Annex A: Thunderstorm) tailored to the Tribal Nation. This structure allows users to navigate the full EOP and work on individual sections independently. The central editing pane contains the active content of the selected chapter or annex. Users may directly edit the text, highlight specific passages, and submit targeted prompts to revise or expand selected content. The system also supports generating new paragraphs on demand to address gaps or updates within a section. The right-hand panel presents a conversational interface that displays text generated by the underlying AI model in response to user prompts.

To further streamline document creation, the system includes a set of pre-defined, ready-to-use prompts accessible through a “Default” option, as shown in Figure 2. These prompts enable users to generate an initial draft of an entire chapter with a single interaction, which can then be reviewed and refined by the user. One example of the predefined prompt was attached in Appendix A. For each chapter, the system presents structured, guided prompts designed to elicit locally grounded information. These prompts are formulated as short questions aligned with FEMA guidance but intentionally open-ended, allowing users to describe Tribal roles, resources, governance structures, mutual aid relationships, and culturally specific practices. Users may modify prompts, add contextual details, or iteratively refine generated text. This design ensures that AI-generated drafts conform to required documentation formats without imposing generic or externally defined assumptions.

To address data sovereignty and confidentiality concerns, all AI computation is performed using locally hosted large language models (LLMs). The backend model for the system is gpt-oss-120b, deployed on secure university-managed infrastructure. No data is transmitted to external cloud services; all user inputs, generated content, and model inference remain within local servers. This architecture ensures that sensitive emergency planning information is stored and processed entirely within institutional and Tribal governance boundaries.

3.3. Evaluation and Data Analysis

This study employed two qualitative user studies to evaluate the usability, applicability, and perceived value of AI Inscriptor for Tribal Emergency Operations Plan (EOP) development. Together, these studies examined practitioner interaction with the system in both focused, in-depth settings and broader, community-centered discussions.

3.3.1. Study 1: Expert Interviews

In the first study, three semi-structured interviews were conducted with Tribal emergency management experts, including a Tribal secretary, a Tribal judge, and a Tribal emergency manager. Each interview session was conducted remotely via Zoom and lasted approximately 60 minutes with one participant and two members from the research team. Participants were invited to interact directly with AI Inscriptor and complete selected EOP chapters and hazard-specific annexes using the system. During the sessions, participants engaged in iterative co-writing tasks, generating and refining draft content for core EOP sections such as Organization and Responsibilities, Concept of Operations, and Recovery. The research team took part in the demonstration and provided guidance on how to use the application. The interviews explored participants' prior experiences with EOP development, FEMA documentation processes, and administrative challenges during disaster response and recovery. After the demonstration and hands on experience, participants were also asked to reflect on their experience using the system, including perceptions of usability, trustworthiness, accuracy, accountability, and the appropriateness of AI assistance in emergency planning and documentation. With the informed consent of the participants, audio recordings were obtained to document their questions, reactions, and experiential feedback concerning the AI tool.

3.3.2. Study 2: Panel Discussion at NCAI 2025

The second study was conducted as a panel style presentation and discussion at the 2025 National Congress of American Indians (NCAI) conference. The 90 minute session introduced AI Inscriptor to approximately 30 participants, including tribal leaders, emergency managers, planners, and policy professionals, representing 19 different tribal nations and stake-holders. The panel included a live system demonstration followed by facilitated discussion and participant feedback on the system's potential applications, benefits, and concerns. A QR code was incorporated into the demonstration to enable participants to test the mobile application. The app was intentionally streamlined to support a single click interaction, allowing researchers to assess whether an AI model could automatically produce an entire chapter of the EOP while minimizing the learning curve for users. Consequently, participants were able to select a predefined prompt, generate a chapter with one tap, and review the output directly on their smartphones. Our system recorded about 50 requests (usages) from the participants. This study emphasized collective reflection and peer discussion. Audio recordings were collected with participant consent to capture questions, reactions, and shared experiences related to emergency planning, documentation burden, and the role of AI in tribal governance contexts.

3.3.3. Data Analysis

Data from both studies—including interview transcripts, detailed session notes, and panel recordings—were analyzed using thematic analysis. This approach aligns with usability assessment practices, particularly in its use of small, qualitative samples and its focus on user-centered evaluation. While grounded in usability evaluation, our approach extends beyond traditional dimensions such as effectiveness and efficiency to also consider factors such as trust, data protection, and contextual relevance, which are critical in tribal emergency management settings. Two researchers independently reviewed the data and collaboratively developed a set of themes related to documentation burden, system usability, trust and accountability, alignment with FEMA requirements, and the preservation of Tribal values. An AI-supported qualitative analysis workflow was used to assist with coding and theme identification, with all interpretations reviewed and validated by the research team (Zhang et al., 2025). Findings from this analysis informed the evaluation of AI Inscriptor's effectiveness as a co-writing tool and identified limitations, such as sensitivity to data used and challenges in the tribal context. Because of privacy considerations, we did not link any individual quotations to specific participants. Instead, the qualitative findings were synthesized and presented as a collective perspective derived from the aggregated interview data.

3.4. Ethics and Researcher Positionality

We recognize that the social, cultural, and institutional identities of its members shape both the interaction with participants and the interpretation of findings. To ensure culturally grounded and ethically responsible engagement, we formed collaborative partnerships with three university faculty and staff members who are enrolled Tribal citizens and co-authors of this manuscript; two of these partners belong to the Omaha Tribe. These alliances were crucial for participant recruitment and for guaranteeing that the study design, data collection, and analysis were informed by Tribal perspectives. We also acknowledge the team's racial and cultural diversity, including

authors of Asian descent, and we remained vigilant to racial dynamics throughout the interviews, especially when participants recounted experiences tied to colonial history and governance.

Participant recruitment and data collection were carried out with careful attention to the ethical complexities of working with communities that may experience environmental and institutional vulnerabilities. Participation was entirely voluntary, and demographic information was gathered only with explicit consent, resulting in intentionally incomplete demographic fields. All interview and panel data were anonymized, stored securely, and used solely for research purposes. The study protocol received approval from the university's Institutional Review Board (IRB). Moreover, the panel based evaluation proceeded only after obtaining formal approval of support from the NCAI organizer. These approvals highlight the centrality of Tribal governance, consent, and trust in research collaborations and demonstrate the ongoing willingness of Tribal governments to engage in this work under formal authorization.

4. Findings

4.1. Study 1: Tribal Experts

Our study revealed three interrelated themes that characterize how AI-supported emergency planning tools are perceived and evaluated in tribal and emergency management contexts. Participants emphasized that usefulness extends beyond drafting speed to include how effectively the tool scaffolds planning work, accommodates local realities, and satisfies requirements for trust and cultural legitimacy. These themes collectively explain the conditions under which such tools are likely to be adopted and sustained in practice.

4.1.1. Theme 1.1: AI as a accelerator could reduce burden

Participants consistently described the tool as significantly accelerating the production of Emergency Operations Plans (EOPs), particularly by drafting the documents which were described as very labor-intensive and requiring extensive time investments often spanning many months or longer. With help of AI it was described as achievable within substantially shorter timeframes. For example, one participant noted that work on EOPs that might otherwise take *"like, a year or two years"* could instead be completed in a matter of weeks or months, emphasizing the dramatic shift in planning timelines. it was noted having something like a draft to start working on was really a big time saver.

However, participants did not evaluate the tool solely on speed. Instead, they emphasized its role as a scaffolded AI co-writer that guides users through the structure, content, and expectations of formal emergency planning documents. Participants highlighted that many practitioners possess deep operational knowledge but lack familiarity with how to translate that knowledge into FEMA-aligned or EOP-ready documentation. As one participant explained, practitioners *"know what needs to be done in their head but they just don't know how to put it into the documents."* In this sense, the tool was valued for helping users understand *"what belongs where"* and for providing structured prompts and sections that reduce uncertainty about required content.

Participants further emphasized that the tool's value lies in its ability to support operational readiness, not just narrative writing. Annexes, templates, and forms were repeatedly identified as the most challenging and most consequential components of emergency planning. Participants noted that plans often expand significantly because *"you can put annex for everything"*, yet these annexes are critical during actual incidents. The ability to generate, extract, print, and sign documents—such as mutual aid agreements or memoranda of understanding—during an emergency was described as a key requirement. One participant explained the need to be able to *"open it up... take that form out... and then have them sign it" in the middle of a disaster.*

4.1.2. Theme 1.2: Trust, legitimacy, and cultural tribal contexts

Across interviews, participants consistently emphasized that trust is a foundational requirement for adoption. Concerns about AI-generated content *"making up information"* emerged as a central barrier, particularly when participants compared the tool to general-purpose AI systems. Participants expressed concern that if content must be extensively verified, the efficiency gains of AI-assisted drafting are undermined. As one participant explained, without confidence in accuracy, users are forced to *"second-guess it, or have to go and try and verify all that information"*, effectively shifting time savings into verification labor. Trust was closely linked to perceptions of reputability and governance. Participants indicated greater confidence in tools that are customized, validated, or associated with trusted institutions, rather than generic or opaque systems. The perception that a tool comes from a *"reputable source"* was seen as critical to both accuracy and acceptance, particularly in professional and governmental contexts.

Cultural and linguistic correctness further shaped trust, especially in tribal settings. Participants expressed strong interest in Indigenous language support and viewed such capabilities as potentially valuable for culturally appropriate planning and communication. However, they also emphasized that correctness is non-negotiable. Experiences with AI-generated language that *"sounded right"* but was *"not written right"* underscored the risk of undermining credibility if linguistic outputs are not validated by fluent community members. These concerns position language support as a trust-sensitive feature requiring careful oversight rather than an optional enhancement.

4.1.3. Theme 1.3: System Usability

Although participants generally described the tool as easy to use, deeper analysis revealed that usability was understood as contextual and relational, rather than a fixed property of the interface. Participants universally expected that AI-generated content would require community localization and expressed little concern if they are fixable. Minor inaccuracies in place names or assumptions about resources were treated as acceptable, as long as users could easily *"go in and change… names"* or adjust details to reflect local conditions. In this way, localization was seen as a normal part of the planning process rather than a flaw in the system.

At the same time, participants emphasized that usability depends heavily on support mechanisms that help users keep going when they encounter obstacles. Participants described themselves—or anticipated end users—as having limited technical confidence. For these users, initial onboarding and clear instructions were described as essential. One participant noted that while the tool became intuitive after some guidance, they *"wouldn't know how to do it without some instruction."* This highlights that perceived ease of use is contingent on early-stage support rather than immediate intuitiveness.

Participants also anticipated challenges for users who are older, less technologically experienced, or unfamiliar with certain devices or operating systems. Comments such as *"Not used to an Apple computer"* and questions like *"How do I scroll up now?"* illustrate how seemingly minor interface or device issues can disrupt use. To mitigate these risks, participants suggested the need for simple, embedded assistance—such as a *"click here"* option that provides the next step when users are stuck—so that frustration does not lead to abandonment.

4.2. Study 2: Tribal Panel on AI Co-Writing

The analysis of the panel discussion identified three interrelated themes that capture how AI is understood and evaluated from a national cross tribal perspective. Across all themes, AI was not treated as a neutral innovation. Instead, it was assessed as a governance intervention that intersects with sovereignty, labor, environmental responsibility, and long-term tribal capacity. Acceptance of AI was consistently conditional and shaped by how well the technology aligned with Indigenous values, practical realities, and historical experience.

4.2.1. Theme 2.1: AI, Sovereignty, and Control of Knowledge

AI was consistently framed through the lens of tribal sovereignty. The central issue was not whether AI is powerful or efficient, but who controls it, who benefits from its use, and who retains decision-making authority. AI was viewed as broadly acceptable only when positioned as a supportive, clerical tool that assists with drafting or organizing EOP documents while leaving all substantive judgment, prioritization, and decision-making with tribal authorities. There was a shared emphasis that tribes already possess the knowledge required to respond to emergencies, even if that knowledge is not recognized or validated by federal systems. As one participant noted, *"We already know what to do. It's an innate ability… but FEMA doesn't buy that."* Within this framing, AI was understood as a potential translator between tribal knowledge and federal bureaucratic requirements rather than a replacement for tribal expertise.

The use of AI to generate policies or determine appropriate actions for communities was explicitly rejected. Participants emphasized that AI should not displace human authority or responsibility, noting that *"It does not create policies. The final decision needs to be on our hands as humans, as users,"* and describing AI instead as *"more like a secretary helping you draft documents. You stay in charge."* Control over data was identified as a core dimension of sovereignty. Trust in AI systems was conditional on clearly articulated data governance aligned with tribal authority. Local hosting, tribal ownership of outputs, and strict limits on data sharing were treated as baseline requirements rather than optional safeguards. As participants emphasized, *"All the data stays local. The tribe owns the database and the documents generated,"* and *"We're not helping big companies train their models."*

Beyond data storage, participants raised concerns about epistemic sovereignty. Even when data are kept local, AI models trained on dominant knowledge systems may embed non-Indigenous values and assumptions. This raised questions about whether Indigenous ways of knowing could

be meaningfully incorporated without exposing sensitive cultural information. As one participant observed, *"I don't see how cultural considerations are possible without training it on sensitive cultural information first."* As a result, sovereignty was framed not as a one-time technical safeguard but as an ongoing governance practice requiring continuous review, consent, and accountability. Trust was strengthened by explicit assurances that tribes could opt out or discontinue use at any time, reinforcing that participation must remain voluntary and reversible.

4.2.2. Theme 2.2: AI, Labor Burden, and the Risk of Extraction

A second theme focused on capacity constraints, workload, and historical patterns of extraction. EOP development was described as a labor-intensive process that often falls to a single individual managing multiple responsibilities. In this context, AI was viewed as potentially useful for reducing administrative burden, particularly for tribes without existing FEMA-aligned templates. Supportive perspectives emphasized AI's ability to accelerate early drafting and reduce reliance on external consultants. Participants noted that AI could *"give an outline and shorten the amount of time spent writing and researching,"* and that reliance on consultants often results in plans that community members do not fully understand: *"Tribes end up hiring consultants, and then nobody knows what's in that document."* From this perspective, AI was seen as a way to keep EOP development within the community and strengthen internal familiarity with emergency plans.

At the same time, concerns were raised that AI could shift rather than reduce labor. Reviewing, correcting, and validating AI-generated content was viewed as potentially increasing workload for already overextended staff. Participants questioned whether AI would ultimately create *"double work,"* noting that *"Now I have to go back and recheck what the machine is outputting,"* and expressing concern that *"If I'm also training the model, that's more labor, not less."* These concerns were closely tied to historical anxieties about extraction. Participants cautioned against repeating patterns in which tribal knowledge, labor, and participation are leveraged for research or innovation without proportional benefit to the community. Warnings included concerns such as, *"Don't shelf it. Don't extract linguistic capital and then reshape it for what you want,"* and skepticism about how AI initiatives differ from prior experiences in which outsiders define what is *"good"* for tribes.

Within this theme, trust depended not only on claims of efficiency but on transparency, accountability, and demonstrated reductions in burden. AI was evaluated based on whether it meaningfully supported tribal capacity without introducing new forms of unpaid labor, surveillance, or dependency. While perspectives varied, there was shared agreement that AI should strengthen—not substitute for—tribal expertise and institutional capacity.

4.2.3. Theme 2.3: Environmental Responsibility, Scale, and Root Causes

The third theme centered on the environmental, relational, and long-term implications of AI adoption. Technology was not evaluated solely on operational efficiency but in relation to land, water, energy use, and responsibilities to future generations.

Participants expressed concern that AI systems, particularly when scaled, may contribute to environmental degradation that exacerbates the very disasters tribes are working to manage. As one participant stated, *"We're almost creating a disaster long-term to address short-term convenience."* Others pointed to the material costs of AI infrastructure, noting that *"Millions of gallons of water are used to cool servers, and that water doesn't go back into the hydrological cycle."*

These concerns extended beyond the specific system demonstrated. Participants emphasized that scale matters and cautioned that even small, localized systems could have significant cumulative impacts when replicated widely: *"I'm not worried about one model. I'm worried about a million tiny models."* Environmental costs were described as frequently obscured in technology discussions, particularly through abstractions such as *"the cloud."*

This theme also intersected with questions about root causes. Participants questioned whether investment in AI might divert attention and resources away from more fundamental needs, such as training tribal members, strengthening emergency management programs, and addressing inequities in federal funding processes. As participants observed, *"The root cause is that we need resources to train our own people,"* and asked, *"What's the next thing after emergency management?"* Despite these concerns, AI was not categorically rejected. Instead, participants emphasized the need for due diligence, including environmental impact assessments, transparent cost–benefit analyses, and tribal governance processes to evaluate tradeoffs. As one participant noted, *"You can't make an informed decision if you don't have the environmental impact information."*

Across this theme, AI was framed not simply as a technical tool but as an ethical and relational choice. Decisions about adoption were expected to account for cumulative impacts, long-term consequences, and Indigenous relationships to place, rather than focusing narrowly on short-term administrative gains.

5. Discussion

This study employed a qualitative, design-oriented methodology that combined document analysis and system design to develop an AI-assisted co-writing tool for Tribal Emergency Operations Planning (EOP). FEMA Tribal emergency documentation requirements and a Tribal Emergency Operations Plan from the Omaha Tribe of Nebraska were analyzed to identify mandatory federal compliance elements, documentation burdens, and the ways federal guidance is operationalized within Tribal governance structures. Insights from this analysis informed the design of AI Inscriptor, a configurable AI-enabled writing platform deployed as a decision-support tool rather than an automated planning system. A FEMA-aligned EOP template was embedded within the platform, enabling emergency managers to generate, edit, and refine chapter-specific content through structured, guided prompts that elicit locally grounded information. The system supports iterative human-in-the-loop drafting while ensuring alignment with required documentation formats and preserving Tribal context, governance, and decision-making practices.

We conducted semi-structured interviews with tribal emergency management professionals, performed hands on testing of the AI Inscriptor platform, and facilitated a panel discussion at the National Congress of American Indians (NCAI). Across both studies, findings show that AI-supported co-writing tools are valued not simply for accelerating document production, but for how well they scaffold emergency planning work, translate operational knowledge into FEMA-aligned documentation, and support real response workflows. Tribal experts emphasized that having a structured draft dramatically reduces the time and cognitive burden of EOP development, particularly for labor-intensive annexes and forms that are critical during emergencies. However, usefulness was consistently defined in relational terms: tools must guide users through "what belongs where," remain editable and localizable, and integrate into existing planning practices rather than replacing them. Ease of use depended on onboarding, recoverability from errors, and support for users with varying technical confidence, underscoring that usability in this context is inseparable from workflow continuity and human judgment.

At the same time, adoption was strongly conditioned by trust, sovereignty, and long-term accountability. Participants across interviews and the national panel framed AI not as a neutral efficiency tool but as a governance intervention whose legitimacy depends on tribal control over data, decision-making, and knowledge. AI was broadly acceptable only when positioned as a clerical assistant—supporting drafting and organization while leaving all substantive authority with tribal governments. Concerns about accuracy, cultural and linguistic correctness, labor shifting, environmental costs, and historical patterns of extraction shaped cautious but engaged responses. Together, these findings suggest that AI tools for emergency planning must be locally governed, transparent, environmentally responsible, and explicitly designed to strengthen—rather than displace—tribal expertise and institutional capacity. This further suggests that usability in this context extends beyond traditional dimensions such as effectiveness and efficiency to include factors such as trust, data protection, and contextual relevance, which are critical in tribal emergency management settings

Design implications for refining AI co-writing in Tribal emergency management. Our findings suggest that AI co-writing tools should be designed less as "faster writing" systems and more as end-to-end planning scaffolds that help practitioners translate operational knowledge into FEMA-aligned structure while remaining easy to localize. Concretely, this implies (1) prioritizing section-level guidance with prompt libraries tuned to annexes, forms, and incident-ready artifacts (e.g., mutual aid/MOU templates that can be generated, exported, printed, and signed during response), (2) supporting rapid verification and editability through features like inline source cues, "needs-local-input" placeholders, and checklist-style validation for high-risk claims to prevent "made-up" information from shifting burden to users, and (3) building usability for uneven technical confidence via lightweight onboarding, embedded "next step" help, undo/version history, and clear save/export pathways. Because linguistic and cultural correctness is trust-sensitive, any Indigenous language functionality should be treated as human-verified workflow support (e.g., community review flags, optional glossaries, and "approve before use" gates).

Design implications for Tribal governance workflows and environmental impact. Participants evaluated AI as a governance intervention; therefore, beneficial deployment requires a tribally legible governance workflow that makes control explicit and reversible. Tools should include role-based permissions (drafting vs. review vs. approval), auditable change logs, and "human decision points" that formalize where Tribal authorities confirm, revise, and sign off—reinforcing AI as clerical support rather than an authoritative planner. Data sovereignty expectations also imply default local hosting, clear statements of ownership over prompts/outputs, opt-out/termination procedures, and policies that prohibit upstream model training without Tribal consent. Finally, to address environmental concerns and improve stakeholder communication, systems should provide an impact disclosure layer: a plain-language dashboard that reports approximate energy use per session, server utilization, and cooling/water implications where applicable, alongside mitigation options (e.g., run smaller models for routine drafting, batch generation

windows, sleep/idle settings, and governance rules limiting unnecessary generation). Presenting these tradeoffs transparently—framed in terms of land, resources, and long-term responsibility—supports informed Tribal decision-making about whether and how AI use aligns with community priorities and capacity-building goals.

Scalability and Practical Implications for Tribal Emergency Planning. The scalability of the proposed tool to other tribal nations is promising, but requires further study. In this work, we use a developed LLM-based system, which relies on pre-training and therefore benefits from access to sufficient and contextually relevant data. As the system is scaled, additional locally collected data could support the development of tribe-specific models that better reflect local governance structures, practices, and knowledge systems. In terms of practical use, the tool appears particularly valuable in contexts where existing Emergency Operations Plans (EOPs) are limited, unstructured, or not regularly updated. In such settings, the system can assist by generating a structured initial draft that emergency personnel can review, revise, and adapt. While human oversight remains essential, this approach can reduce the burden of starting from scratch and support more consistent and organized documentation.

6. Conclusion

This paper demonstrates that AI-assisted co-writing can meaningfully support Tribal Emergency Operations Planning when designed and governed as a collaborative, locally controlled infrastructure rather than a purely efficiency-driven technology. Through document analysis, system design, and qualitative evaluation with Tribal practitioners and leaders, we show that tools like AI Inscriptor can reduce documentation burden, scaffold complex FEMA-aligned planning tasks, and help translate operational knowledge into usable emergency plans—while keeping Tribal experts firmly in control. At the same time, our findings highlighted that adoption hinges on trust, sovereignty, accountability, and transparency, including clear governance workflows and honest communication about environmental impacts. Together, this study contributes design and governance insights for responsible AI integration in Tribal emergency management and, more broadly, for public-sector AI systems that must balance regulatory compliance, local knowledge, and long-term community capacity.

7. Contributor Statement

Chun-Hua Tsai: Conceptualization, Methodology, Investigation, Formal Analysis, Data Curation, Writing – Original Draft, Writing – Review and Editing, Supervision.
Taranum Bano: Investigation, Data Curation, Writing – Review and Editing.
Yu-Che Chen: Investigation,Data Curation, Resources
Ed Zendejas: Investigation,Data Curation, Resources
Sarah Crafka: Investigation,Data Curatiom, Resources

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Use of AI

AI tools were used minimally for minor language and editing support. All content was developed and verified by the authors.

Conflict of Interest

The authors declare that there is no conflict of interest.

Data/Software Access Statement

The data and software that support the findings of this study are available on request from the corresponding author.

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A. Pre-defined Prompt

The following example prompt is designed to generate a full chapter titled *Purpose and Scope of the Plan*. It provides clear instructions for users to develop a comprehensive chapter together with an illustrative policy excerpt.

> Generate a section titled **1.1 PURPOSE AND SCOPE OF THE PLAN** that follows this exact format and professional tone.
> Keep all structure and flow same — only make the following parts editable: - [Tribe or Organization Name] - [Plan Type] (e.g., Hazard Mitigation Plan, Emergency Operations Plan) - [State or Region] - [Relevant Federal Acts or Statutes] (e.g., Disaster Mitigation Act of 2000, Robert T. Stafford Act, Section 322) - [Grant or Program Name] (e.g., Hazard Mitigation Plan Grant - HMPG) - [Federal Agency Name] (e.g., FEMA)
> **Template to follow (auto-fill and rephrase based on the provided values):**
> **1.1 PURPOSE AND SCOPE OF THE PLAN**
> The purpose of the [Tribe or Organization Name]'s [Plan Type] is to provide a comprehensive set of guidelines to reduce the [Tribe or Organization Name]'s vulnerability to disasters. The plan identifies potential risks with appropriate mitigation responses to significantly reduce loss of life, injuries, economic costs, and destruction of natural and cultural resources.
> This plan was developed pursuant to the regulations of the [Relevant Federal Act] ([Abbreviation]) of 2000. The [Act Abbreviation] revises the [Additional Act or Section, e.g., Robert T. Stafford Disaster Relief and Emergency Assistance Act] by adding [Section number, if any], which provides new and revitalized emphasis on hazard mitigation, including new requirements for tribal or local mitigation plans. In order to be eligible for [Grant or Program Name] project funding, the new legislation required each tribe or local entity to have a [Federal Agency Name] approved hazard mitigation plan.

The [Tribe or Organization Name] shall be referred to as the “Tribe” (or “Organization”) for purposes of this document. The Tribe is a federally recognized governing body and implements a constitutional form of government. In recognition of the Tribe’s grantee status with [Federal Agency Name], this Tribal Plan submission is in full compliance with a State Level Plan. The Tribe is committed to comply with all applicable Federal statutes and regulations during the periods for which it receives grant funding, in compliance with [Federal Regulation Reference, e.g., 44 CFR 13.11(c)], and will amend its plan whenever necessary to reflect changes in Tribal or Federal laws and statutes as required in [Federal Regulation Reference, e.g., 44 CFR 13.11(d)]. —
Formatting requirements: - Keep all text in professional FEMA-style paragraph structure (3 paragraphs). - Maintain the numbering and heading style. - Use formal tone and consistent spacing/alignment like an official mitigation plan.